



PROJECT TALA

GEOSPATIAL POVERTY INCIDENCE ESTIMATION

Project TALA: User's Manual

Geospatial Poverty Incidence Estimation

1. Introduction

Project TALA is a geospatial analytics platform designed to estimate poverty incidence across the Philippines using satellite imagery and advanced machine learning architectures. This manual provides a step-by-step guide on how to navigate the system, search for specific municipalities, and generate comprehensive poverty incidence reports.

2. System Navigation Guide

Step 1: System Access and Authentication

To access Project TALA, navigate to the system's URL. You will be greeted by the secure login screen.

- Access the system's URL <https://datamaraws.vercel.app/>
- Enter **admin** as **Username** and **tala2026** as **Password**.
- Click the **Sign in** button to authenticate and access the main dashboard.

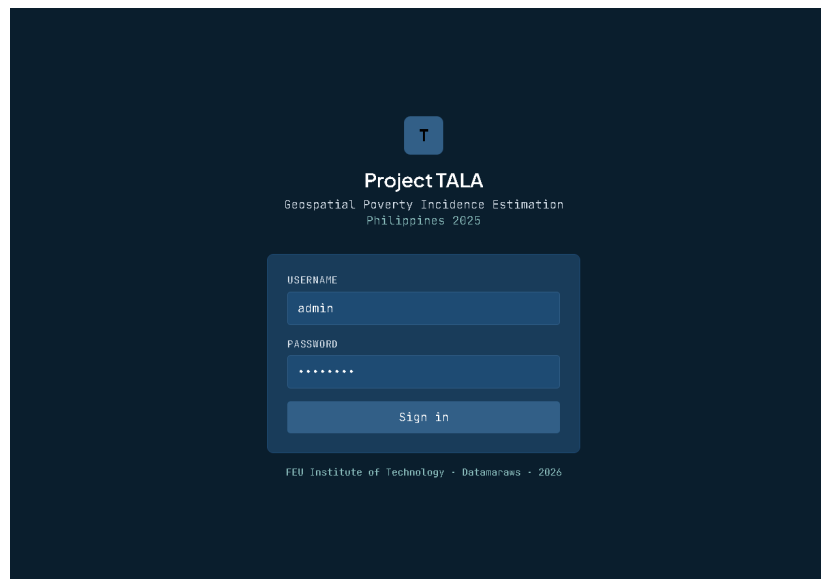


Figure 1: Project TALA Login Interface



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Step 2: Exploring the Main Dashboard

Upon logging in, you will be directed to the main map interface.

- **Heatmap Visualization:** The map displays poverty incidence clusters across the Philippines.
- **Wealth Index Legend:** Located in the bottom left corner, the legend helps you interpret the map colors, ranging from dark orange/red (Poor) to light blue/teal (Wealthy).
- **Navigation:** You can pan across the map by clicking and dragging or zoom in and out using the **plus** (+) or **minus** (-) controls in the top right corner.

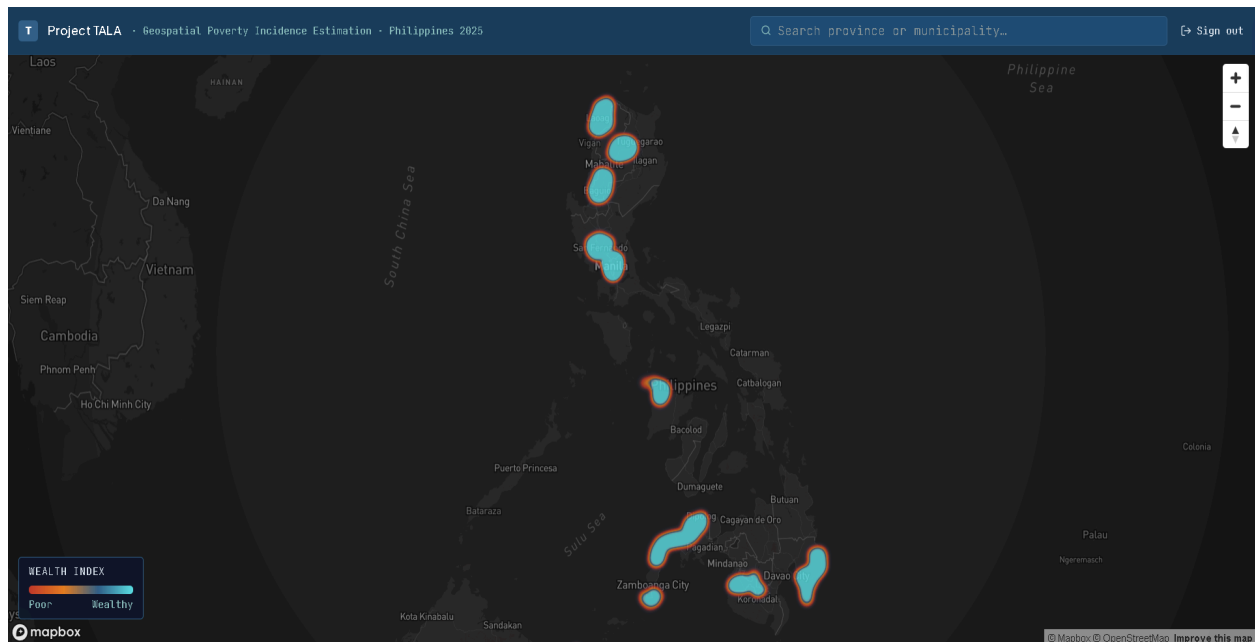


Figure 2: Main Map Dashboard and Wealth Index Legend



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Step 3: Searching for a Specific Location

To analyze a specific province or municipality:

- Locate the **Search bar** at the top right of the screen.
- Type the name of the target area (e.g., "Manila").
- Select the correct result from the dropdown menu (e.g., "City of Manila, Municipality – NCR") to center the map on that specific area.

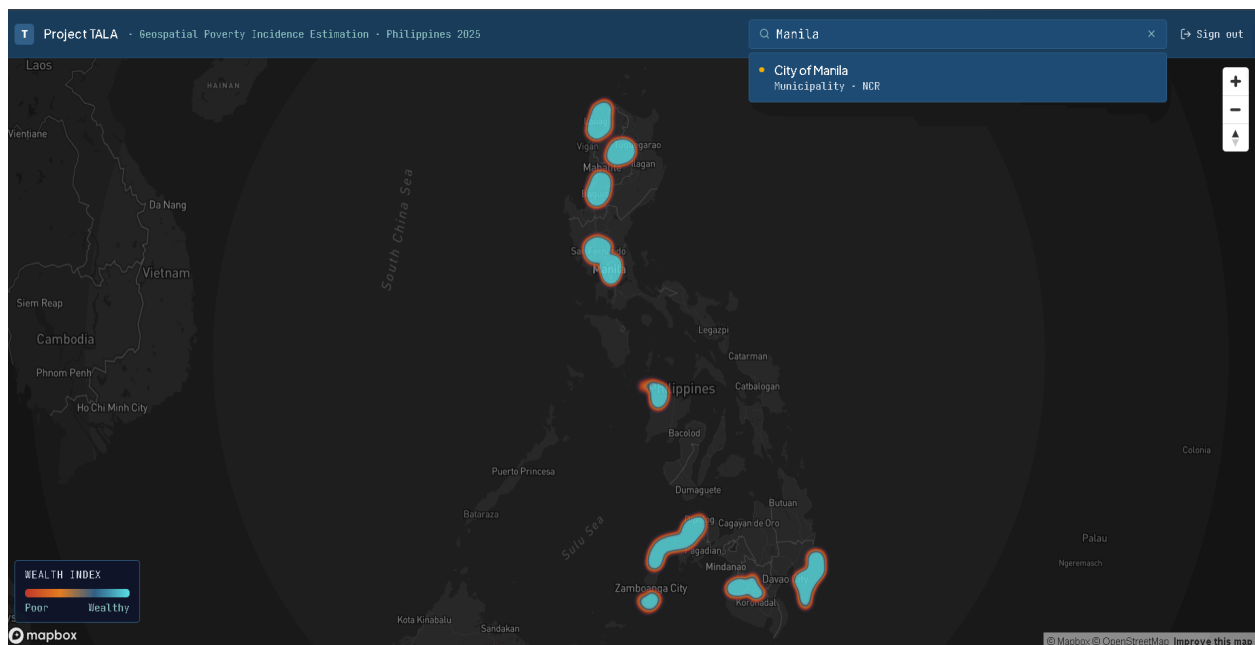


Figure 3: Search Bar Functionality



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Step 4: Viewing Area Summaries and Triggering Reports

Once a municipality is selected, the map will highlight the area's boundaries and display specific data points. A sidebar will also appear on the right side of the screen.

- **Data Overview:** The sidebar provides a quick snapshot of the area, including the Estimated Wealth Index (Mean), Area Summary (Points, VIIRS Mean, Building Density), and DHS Survey validation.
- **Generate Report:** To get a deep dive into the area's data, click the **Generate Report** button at the bottom of the right sidebar. You can also click the download icon next to it to export the summary.

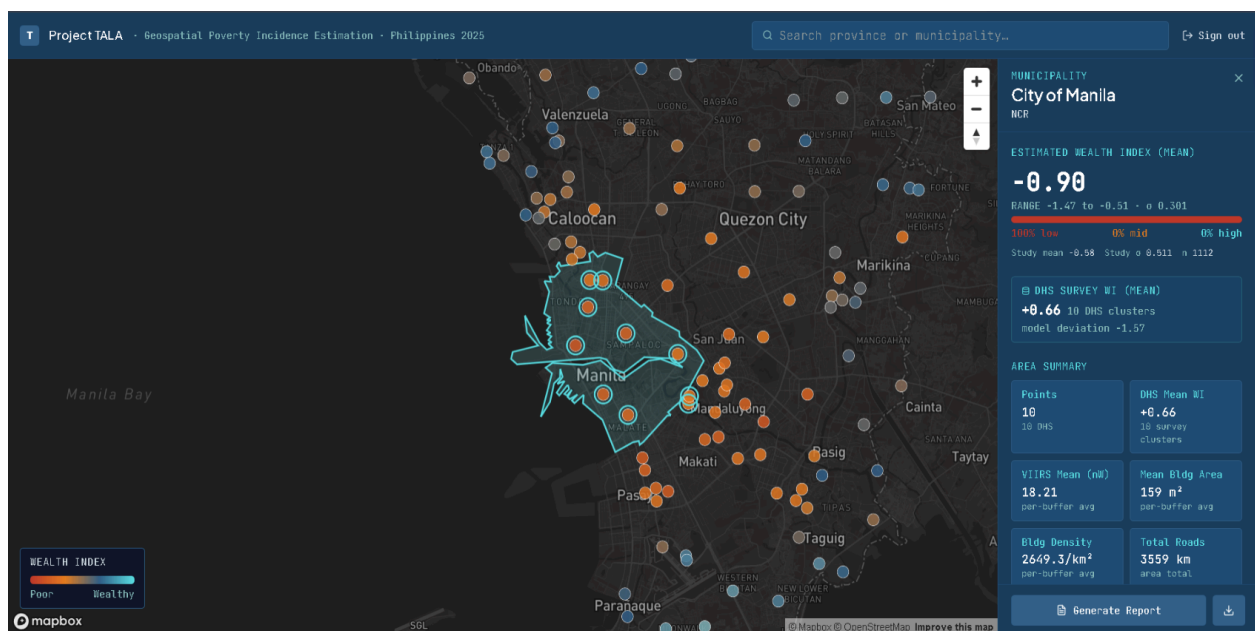


Figure 4: Municipality Overview Sidebar and Generate Report Button



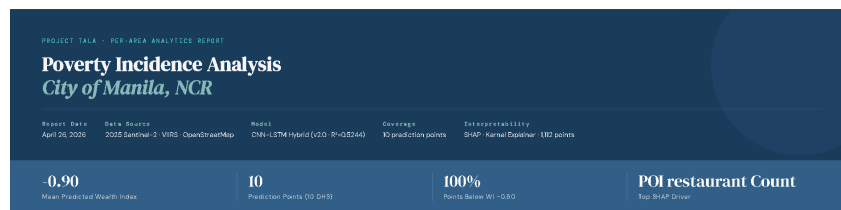
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Step 5: Viewing the Poverty Incidence Analysis Report

Clicking "**Generate Report**" opens a detailed, multi-section analytics page for the selected area. This report can be exported and used for deeper research or policy planning.

A. Area Overview The top section provides the exact model parameters (e.g., CNN-LSTM Hybrid), data sources, and high-level metrics like the Mean Predicted Wealth Index and the strongest SHAP driver (e.g., POI restaurant Count).



Six key performance indicator cards are displayed at the top of the report. Each card shows one summary metric for the selected area. The sub-text below each value places it in national context by showing the study-wide mean and standard deviation. Cards showing a value more than one standard deviation below the national mean are flagged in a warm color to draw attention to underperformance.

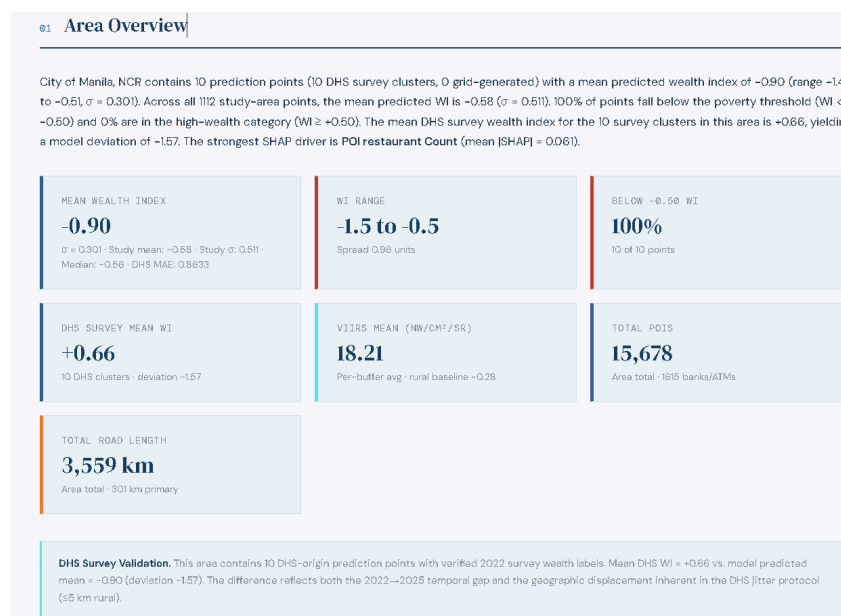


Figure 5: Poverty Incidence Analysis Report Header and Overview



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- **Mean Predicted WI.** The average predicted wealth index across all prediction points in the area. Ranges from approximately -1.7 (very poor) to +0.2 (relatively wealthy) in the current study. Negative values indicate below-average asset wealth relative to the national DHS mean.
- **WI Range and Spread.** The minimum, maximum, and standard deviation of predicted wealth index (WI) within the area. A wide spread indicates high within-area inequality. A narrow spread indicates homogeneous wealth conditions.
- **Percentage Below Poverty Threshold.** Percentage of prediction points with WI below -0.50, which is used as the study-defined low-wealth threshold. A high percentage indicates a predominantly poor area by model estimates.
- **DHS Survey Mean WI.** When DHS-origin clusters are present in the area, the actual mean 2022 DHS Wealth Index from those clusters is shown. This is the ground truth against which model predictions can be validated.
- **Mean VIIRS Radiance.** Average nighttime luminosity within the area in $\text{nW/cm}^2/\text{sr}$. Used as a proxy for electrification and economic activity level.
- **Total Road Length.** Combined length of all road segments within the prediction point buffers of the area, in km.

B. Point-Level Wealth Index Estimates. Scrolling down, the system displays a tabular breakdown of prediction points within the municipality. It compares the Predicted Wealth Index (Pred. WI) against the verified Demographic and Health Surveys (DHS) label, alongside infrastructure data like total Point of Interests (POIs) and VIIRS radiance.



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02 Point-Level Wealth Index Estimates

Up to 20 prediction points within City of Manila, NCR, sorted ascending by predicted wealth index. DHS-origin points show the verified survey wealth index (DHS WI) alongside the model prediction. Grid-generated points show — in the DHS WI column.

POINT ID	MUNICIPALITY	TYPE	SOURCE	PRED. WI	DHS WI	VIIRS (NW)	TOTAL POIS	STATUS
01096	City of Manila	Urban	DHS	-1.47	+0.01	18.05	2043	Low Wealth
01098	City of Manila	Urban	DHS	-1.18	+0.55	18.25	2430	Low Wealth
01094	City of Manila	Urban	DHS	-1.15	+0.58	15.55	1651	Low Wealth
01101	City of Manila	Urban	DHS	-1.10	+0.15	19.85	2134	Low Wealth
01099	City of Manila	Urban	DHS	-0.98	+1.02	20.55	2333	Low Wealth
01097	City of Manila	Urban	DHS	-0.78	+0.94	13.55	912	Low Wealth
01095	City of Manila	Urban	DHS	-0.65	+0.49	13.85	638	Low Wealth
01100	City of Manila	Urban	DHS	-0.63	+1.39	19.27	1076	Low Wealth
01092	City of Manila	Urban	DHS	-0.60	+0.75	21.45	1181	Low Wealth
01073	City of Manila	Urban	DHS	-0.51	+0.75	21.70	1280	Low Wealth

Model note. $R^2 = 0.5244$, Pearson $r = 0.7271$ (5-fold CV, 1,234 DHS clusters). Inference applied to 2025 Sentinel-2 composites. DHS WI = verified 2022 survey label (DHS-origin points only).

Figure 6: Point-Level Wealth Index Data Table



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C. Wealth Index Distribution This visual section helps users understand the spread of data through graphs:

- **Sorted Bar Chart (top 20 prediction points).** A bar chart showing the predicted wealth index for the top twenty prediction points in the area, sorted from the lowest to the highest value. Each bar represents one prediction point. The color of each bar indicates the wealth class: red-orange for low wealth (WI below -0.50), amber for below average (-0.50 to $+0.50$), and teal for high wealth (above $+0.50$). This chart is most useful for identifying the most economically distressed specific locations within an area.
- **VIIRS Median vs Predicted WI Scatter Plot.** A scatter plot of nighttime luminosity (VIIRS_Median, x-axis) against predicted wealth index (y-axis) for up to 80 prediction points in the area. Each point is colored by its province. The upward trend expected from this plot reflects the positive relationship between electrification and wealth. Points that fall far below the trend line (high VIIRS but low WI) may indicate areas with commercial lighting but limited household wealth, such as industrial zones or transit corridors. Points above the trend line may indicate residential areas with high asset wealth but lower nighttime lighting penetration.
- **WI Histogram.** A frequency histogram showing the distribution of predicted WI values across all prediction points in the area, using 0.25-unit bins. The shape of the histogram reveals the overall wealth profile. A histogram skewed strongly to the left (toward negative values) indicates a predominantly poor area. A roughly symmetric histogram centered near zero indicates a mixed-wealth area. The red dashed line marks the low-wealth threshold at $WI = -0.50$.
- **DHS WI vs Estimated WI Grouped Bar Chart:** A grouped bar chart comparing the mean DHS Survey WI (teal bar) against the mean Predicted WI (navy bar) for each province in the study, using national data. This chart provides a study-wide calibration view. The closer the two bars are for each province, the better the model calibration.



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03 Wealth Index Distribution



Figure 7: Wealth Index Distribution Charts

D. Infrastructure Profile The final section provides environmental and structural context for the AI's predictions:

- Infrastructure Radar:** A hexagonal radar chart with eight axes representing key infrastructure indicators: Total Road Length, Total Buildings, Building Density, Financial Services (bank and ATM count), Education (school building count), Health (hospital and pharmacy count), Total POIs, and VIIRS Median. Each axis is normalized relative to the study-wide range for that indicator, so a score near the outer edge means the area ranks highly among all study provinces. A score near the center means the area ranks poorly. The shape of the polygon reveals the area's infrastructure profile at a



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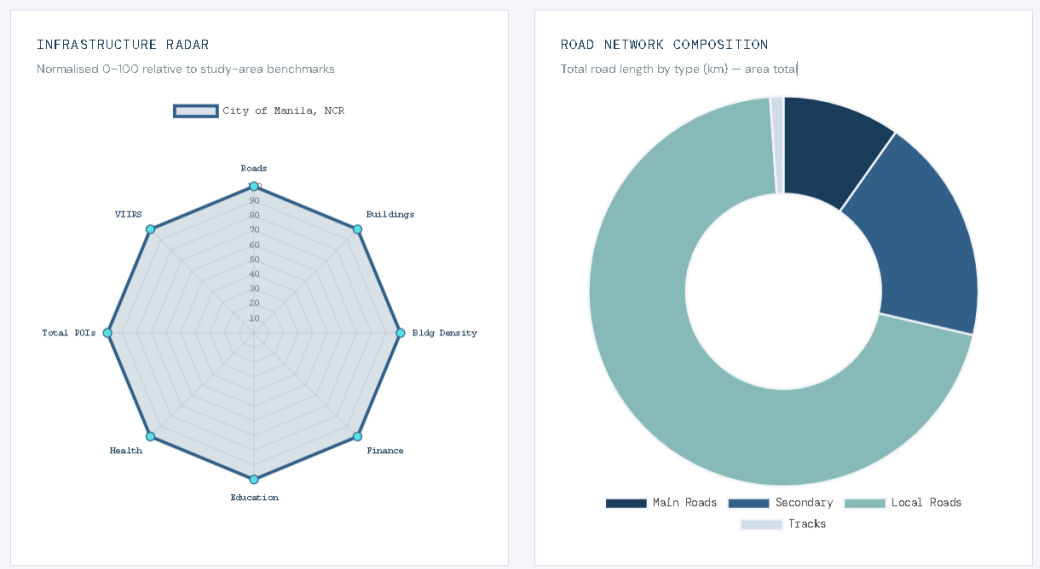
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glance. A balanced polygon that is uniformly small indicates uniformly poor infrastructure. An uneven polygon with spikes on some axes indicates uneven development.

- **Road Network & Landuse Composition:** Doughnut charts showing the proportional breakdown of road types: main roads, secondary roads, local roads, and tracks; and the proportional composition of landuse by class: agricultural, forest, residential, commercial, and industrial.
- **Building Type Count Bar Chart.** A vertical bar chart showing the total count of buildings by type: residential, commercial, industrial, school, and hospital.
- **Mean Building Area by Type Bar Chart.** A vertical bar chart showing the average footprint area in square meters for each building type. This chart measures infrastructure quality rather than quantity.
- **Horizontal Service Infrastructure Bar Chart.** A horizontal bar chart showing the count of eight service and facility categories: banks and ATMs, schools, restaurants, pharmacies, markets, health facilities, and government and police offices.

04 Infrastructure Profile

Infrastructure metrics for City of Manila, NCR. Area totals represent the sum across all prediction point buffers (roads, buildings, POIs, landuse). Per-buffer averages (VIIRS, mean building area, building density, proportions) reflect the typical value for a single buffer in this area.





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Figure 8: Infrastructure Profile and Composition Charts

E. Municipal Breakdown (For Province-level Reports)

- Horizontal Ranked Bar Chart.** A horizontal bar chart ranking all municipalities within the province by mean predicted wealth index, from lowest to highest. This chart directly answers the question of which municipalities within the province are the most and least wealthy by model estimates.
- Municipal Summary Table.** A table accompanying the bar chart listing each municipality with its mean predicted WI, total number of prediction points, and predominant wealth class.

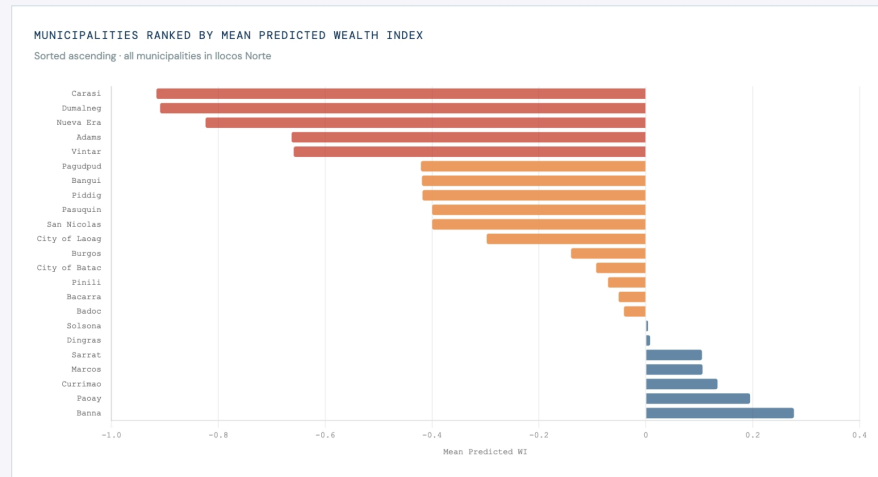


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05 Municipal Breakdown

23 municipalities within Ilocos Norte ranked by mean predicted wealth index. 5 municipalities average below the poverty threshold (WI < -0.50). Carasi records the lowest mean WI (-0.92) and Banna the highest (+0.28).



MUNICIPALITY	POINTS	MEAN WI	% BELOW -0.50	STATUS
Carasi	6	-0.92	83%	Low Wealth
Dumalneg	1	-0.91	100%	Low Wealth
Nueva Era	23	-0.82	74%	Low Wealth
Adams	3	-0.66	67%	Low Wealth
Vintar	21	-0.66	71%	Low Wealth
Pagudpud	7	-0.42	57%	Below Average
Bangui	4	-0.42	25%	Below Average
Piddig	4	-0.42	50%	Below Average
Pasuguin	7	-0.40	29%	Below Average
San Nicolas	2	-0.40	0%	Below Average
City of Laoag	6	-0.30	17%	Below Average
Burgos	4	-0.14	25%	Below Average
City of Batac	6	-0.09	0%	Below Average
Pinili	2	-0.07	0%	Below Average
Bacarra	2	-0.05	0%	Below Average
Badoc	3	-0.04	0%	Below Average
Solsona	3	+0.00	0%	Moderate
Dingras	4	+0.01	25%	Moderate
Sarrat	2	+0.10	0%	Moderate
Marcos	2	+0.11	0%	Moderate
Currimao	1	+0.13	0%	Moderate
Paoay	2	+0.20	0%	Moderate
Banna	4	+0.28	0%	Moderate

Figure 8: Municipal Breakdown Data



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E. Interpretability (Data Interpretation): This section visualizes exactly how the CNN-LSTM model arrived at its conclusions for the area. SHAP values represent the average magnitude of each feature's contribution to the predicted wealth index across all prediction points in the area. A positive SHAP direction means higher values of that feature push predictions toward higher wealth. A negative SHAP direction means higher values push predictions toward lower wealth.

- **SHAP Feature Drivers.** A horizontal bar chart showing the top twelve features by their mean absolute SHAP value across all prediction points in the area. Each bar is colored by feature category. A longer bar means the feature has a larger average influence on the predicted wealth index. The chart answers which features are driving the model's predictions for this specific area.
- **SHAP Category Contribution Bar Chart.** A bar chart grouping SHAP contributions by category: Temporal, Satellite, Road, Building, POI, and Landuse. This chart shows the relative importance of each category in aggregate.
- **Ranked Attribution Table.** A tabular listing of all features with their mean absolute SHAP value and feature category.

05 SHAP Feature Attribution

SHAP values quantify the marginal contribution of each feature to wealth index predictions in City of Manila, NCR. Temporal satellite features (LSTM) account for approximately 51.8% of total absolute SHAP contribution per cluster, with static OSM and VIIRS features contributing 48.2%. This near-equal split reflects the expanded 52-feature static branch in the retrained model.





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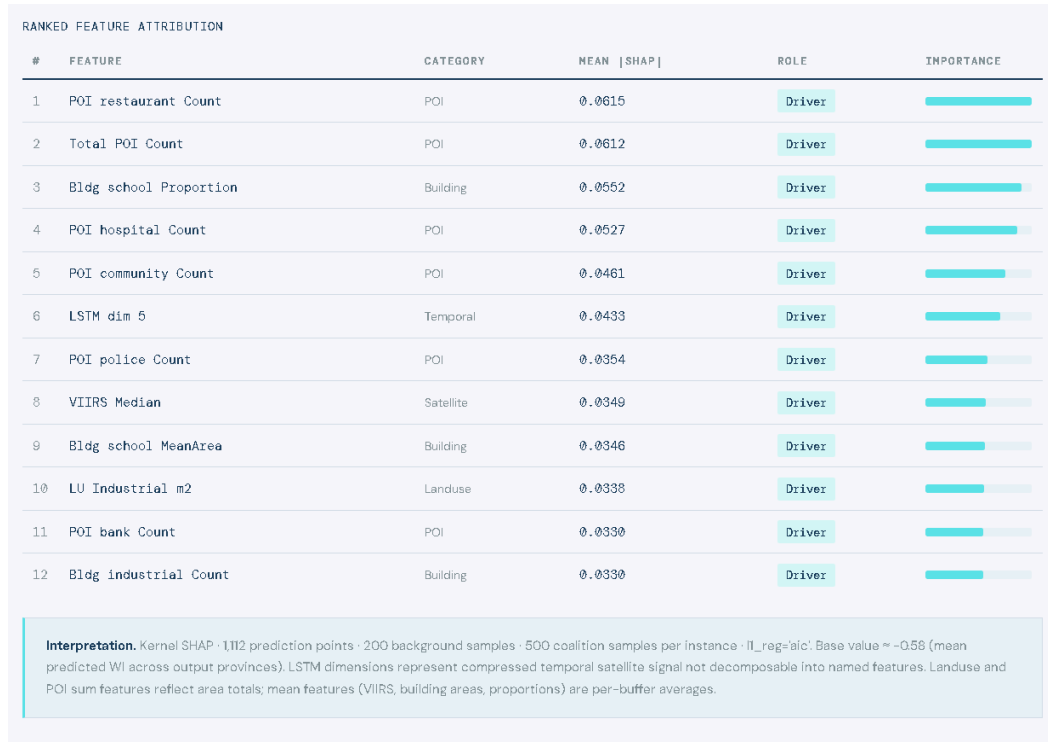


Figure 9: Interpretability and SHAP Data Visualizations

F. Policy Recommendations. Up to four priority recommendation cards are generated from the highest-ranked SHAP features that have mapped policy domains. Each card shows the policy domain, the relevant government agency responsible for action, the observed feature value for the selected area, and a narrative policy implication. Recommendations are ordered by SHAP signal magnitude so that the most influential drivers for this specific area appear first. Cards are specific to the selected area and will differ between provinces and municipalities based on their SHAP profiles.



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07 Policy Recommendations

Derived from SHAP feature attribution for Ilocos Norte. Ranked by SHAP signal magnitude.

PRIORITY 1 • AGRICULTURAL DEVELOPMENT • SHAP 0.095 • DA / DAR / DTI

Agricultural Development Intervention in Ilocos Norte

Agricultural landuse (118 km² total, SHAP = 0.095) is the strongest static predictor in Ilocos Norte. High agricultural coverage with low wealth indicates subsistence farming. DA should prioritise crop diversification, value chain access, and smallholder credit in the lowest-wealth municipalities.

PRIORITY 2 • SERVICE ACCESSIBILITY • SHAP 0.070 • DILG / DTI / LGU

Service Accessibility Intervention in Ilocos Norte

Total service facility count (3609 total POIs in Ilocos Norte, SHAP = 0.070) captures aggregate economic footprint. Low POI density signals limited market access. LGU infrastructure investment and DTI enterprise development should target underserved municipalities.

PRIORITY 3 • EDUCATION INFRASTRUCTURE • SHAP 0.050 • DEPED / CHED

Education Infrastructure Intervention in Ilocos Norte

School building count (9354 total in Ilocos Norte, SHAP = 0.050) is a positive predictor of wealth. DepEd should prioritise construction, repair, and teacher deployment in municipalities where school count falls below the provincial average.

PRIORITY 4 • EDUCATION QUALITY • SHAP 0.050 • DEPED / DPWH

Education Quality Intervention in Ilocos Norte

Mean school building area (144 m² per buffer, SHAP = 0.050) proxies education infrastructure quality in Ilocos Norte. DepEd and DPWH should prioritise structural upgrades and multi-storey classroom construction in municipalities with the smallest average school footprint.

Limitations. Recommendations are derived from a model trained on 2022 DHS data applied to 2025 imagery. Validation against ground-truth surveys and community consultation is required before policy action.

Figure 10: Policy Recommendations

G. National Analysis. This section analyzes all prediction points on a national level to provide context to the localized analysis presented above.

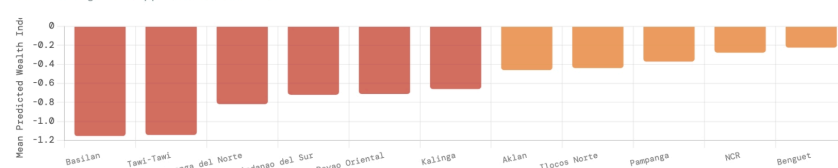
- **All-Province Ranking Bar Chart.** A horizontal bar chart ranking all eleven study provinces by mean predicted WI from lowest to highest.

08 National Analysis

This section ranks all 11 study provinces and 166 municipalities by predicted wealth index. The study covers 1112 prediction points across the 5 poorest provinces, 5 richest provinces, and NCR. Aggregate infrastructure totals are drawn from osm_agg in the provincial data file.

PROVINCES RANKED BY MEAN PREDICTED WEALTH INDEX

Sorted ascending • all 11 study provinces • 2025 inference





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- DHS WI vs Predicted WI (Province Level).** A scatter plot with mean DHS Survey WI on the x-axis and mean predicted WI on the y-axis, one point per province. The diagonal line represents perfect calibration (predicted equals actual). Points above the line indicate provinces where the model over-estimates wealth; points below indicate under-estimation. This is the primary model validation visualization at the province level.
- Provincial Level Comparison Charts (VIIRS, Roads, POIs).** Three bar charts comparing all provinces on their mean VIIRS radiance, total road length, and total POI count.



- Provincial and Municipal Tables.** Tabular summaries of all eleven provinces and the twenty lowest-wealth and twenty highest-wealth municipalities across the study area.



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PROVINCE SUMMARY TABLE

#	PROVINCE	POINTS	MEAN WI	DHS MEAN WI	VIIRS (NW)	TOTAL POIS	TOTAL ROADS	STATUS
1	Basilan	42	-1.15	-1.18	0.47	137	2,864 km	Low Wealth
2	Tawi-Tawi	29	-1.14	-1.20	0.47	196	1,088 km	Low Wealth
3	Zamboanga del Norte	187	-0.82	-0.61	0.45	958	12,427 km	Low Wealth
4	Maguindanao del Sur	83	-0.72	-1.18	0.47	137	7,336 km	Low Wealth
5	Davao Oriental	157	-0.71	-0.43	0.32	918	9,840 km	Low Wealth
6	Kalinga	110	-0.66	-0.15	0.44	378	5,106 km	Low Wealth
7	Aklan	51	-0.46	+0.03	0.50	1,885	6,330 km	Below Average
8	Ilocos Norte	119	-0.44	+0.38	0.49	3,609	13,458 km	Below Average
9	Pampanga	84	-0.37	+0.84	2.44	15,902	44,581 km	Below Average
10	NCR	149	-0.28	+0.78	12.51	120,193	43,082 km	Below Average
11	Benguet	101	-0.22	+0.52	0.61	19,544	14,253 km	Below Average

20 LOWEST-WEALTH MUNICIPALITIES

Municipalities most likely to benefit from targeted poverty interventions

#	MUNICIPALITY	PROVINCE	POINTS	MEAN WI	% LOW WI	STATUS
1	Simunul	Tawi-Tawi	2	-1.92	100%	Low Wealth
2	Datu Piang	Maguindanao del Sur	2	-1.88	100%	Low Wealth
3	Tuburan	Basilan	2	-1.66	100%	Low Wealth
4	Hadji Mohammad Ajul	Basilan	3	-1.60	100%	Low Wealth
5	Al-Barka	Basilan	1	-1.49	100%	Low Wealth
6	Ungkaya Pukan	Basilan	2	-1.37	100%	Low Wealth
7	Sitangkai	Tawi-Tawi	3	-1.36	100%	Low Wealth
8	Sumisip	Basilan	11	-1.35	100%	Low Wealth
9	Tabuan-Lasa	Basilan	1	-1.30	100%	Low Wealth
10	Sibuco	Zamboanga del Norte	4	-1.27	100%	Low Wealth
11	Leon T. Postigo	Zamboanga del Norte	8	-1.26	100%	Low Wealth
12	Pagalungan	Maguindanao del Sur	7	-1.26	100%	Low Wealth
13	Lantawan	Basilan	5	-1.24	100%	Low Wealth
14	Panglima Sugala	Tawi-Tawi	8	-1.22	100%	Low Wealth
15	Sto. Tomas	Pampanga	1	-1.16	100%	Low Wealth
16	Datu Anggal Midtimbang	Maguindanao del Sur	2	-1.16	100%	Low Wealth
17	Baganga	Davao Oriental	26	-1.11	92%	Low Wealth
18	Mangudadatu	Maguindanao del Sur	3	-1.09	100%	Low Wealth
19	Sibutu	Tawi-Tawi	4	-1.08	100%	Low Wealth
20	Bongao	Tawi-Tawi	5	-1.05	60%	Low Wealth

Figure 11a, b, c: National Analysis Section



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Step 6: Sign Out of the Platform

To ensure data security once your analysis is complete, securely close your session.

- Navigate back to the main dashboard or locate the user profile menu.
- Click the **Sign Out** button on the top right corner to terminate your active session and return to the secure login screen.

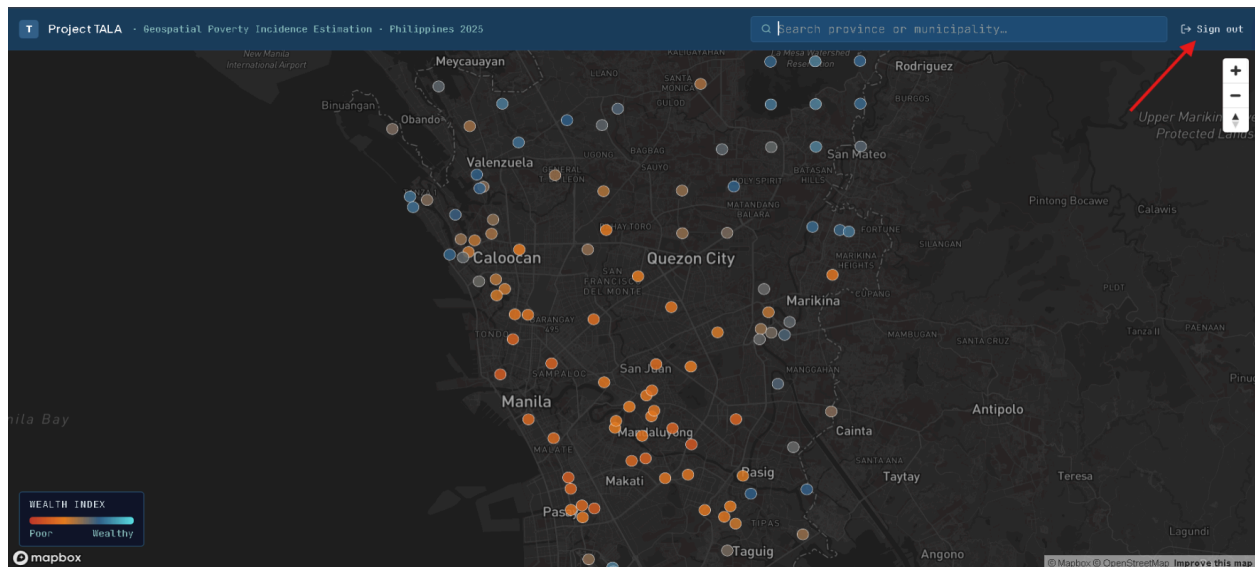


Figure 12: Sign Out Function



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Dynamic/Temporal Data Features

1 Daytime Temporal Satellite Imagery (Sentinel-2)

Source: European Space Agency Sentinel-2 Level-2A Surface Reflectance, accessed via Google Earth Engine.

The daytime satellite imagery is the primary spatial input to the model. For each prediction point, four cloud-filtered median composite images are acquired, one per calendar quarter (Q1: January–March, Q2: April–June, Q3: July–September, Q4: October–December). Each image covers a 10 km x 10 km area centered on the prediction point at 10-meter resolution.

Five spectral bands are used per image:

- B4 (Red, 665 nm)
- B3 (Green, 560 nm)
- B2 (Blue, 490 nm)
- B8 (Near-Infrared / NIR, 842 nm)
- B11 (Short-Wave Infrared / SWIR, 1610 nm)

How It Is Used In The Model

Each quarterly image tile is passed through the VGG16 convolutional neural network to produce a 4,096-dimensional feature vector encoding spatial patterns of land use, building density, road networks, and vegetation. These four quarterly vectors are then processed by the LSTM network, which learns how the spatial patterns change across seasons. The seasonal variation captures signals such as agricultural harvest cycles, dry-season urban heat signatures, and temporal stability of built-up structures that a single image cannot capture.



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Static Features

1 Nighttime Satellite Imagery: VIIRS DNB (VIIRS_Median)

Source: NASA VIIRS Day/Night Band product VNP46A1, accessed via NASA LAADS DAAC.

Unit: nanowatts per square centimeter per steradian (nW/cm²/sr)

VIIRS_Median is the annual median nighttime luminosity value extracted from all available daily VIIRS observations within the calendar year, within a 2 km (urban) or 5 km (rural) buffer around each prediction point. A median reducer is used rather than a mean to suppress transient light sources such as fishing vessel lights, fire events, and atmospheric airglow that produce high-value outliers on individual days.

What It Measures

Nighttime luminosity is produced primarily by artificial light from electricity use. It captures the combined effect of street lighting density, commercial activity (illuminated shopfronts, night markets), industrial operations, and residential electricity access. It is one of the most consistently validated proxies for economic activity at the sub-national level across developing countries.



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2 OSM Road Network Features

Source: OpenStreetMap via Geofabrik Philippines extract, roads layer.

Unit: kilometers (km)

Road features measure the total length of paved and unpaved roads within the prediction point buffer, broken down by road hierarchy. Road access is a fundamental determinant of poverty, as it conditions agricultural commercialization, access to services, and labor market participation.

Total_Road_Length

Sum of all road and path lengths within the buffer in km, including all categories below. The broadest indicator of overall transport connectivity. Areas with higher total road length are more physically connected to economic activity. Highly rural and remote areas typically have very low values.

Main_Roads_Length

Total length of high-order roads: motorways, trunk roads, primary roads, and primary links. These carry inter-city and inter-provincial traffic. Presence of main roads near a cluster indicates proximity to major economic corridors and market access. Urban and peri-urban areas typically have higher values.

Secondary_Roads_Length

Total length of secondary and tertiary roads connecting municipalities to provincial capitals and regional centers. Secondary road presence indicates access to healthcare, education, and markets at the sub-provincial level.

Local_Roads_Length

Total length of residential streets, unclassified roads, living streets, and service roads. These are the internal circulation network of a community. Higher values indicate denser settlement patterns and more active internal economic activity.

Tracks_Length

Total length of unpaved tracks and footpaths. Tracks are the informal transport infrastructure of remote and upland communities. High track length relative to total road length signals a community that relies on informal access rather than paved roads, which is strongly associated with



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geographic isolation and poverty. In the model, high track proportion has a negative SHAP direction, meaning more tracks correlates with lower predicted wealth.

3 OSM Building Features

Source: OpenStreetMap buildings layer.

Building features capture the physical footprint and density of structures within the prediction point buffer. Building density and mean footprint size are structural proxies for urbanization level, construction quality, and investment in the built environment.

Aggregate Building Features

Total_Bldg_Count

Total number of building polygons within the buffer. Directly reflects settlement density. Urban areas have counts in the thousands; remote rural areas may have fewer than ten mapped buildings, partly reflecting OSM mapping completeness.

Total_Bldg_Area (m²)

Sum of footprint areas of all buildings in the buffer. Larger total area indicates denser or larger-scale built structures. Commercial and industrial areas have high total area even at lower building counts.

Mean_Bldg_Area (m², per-buffer average)

Average footprint area per building. A proxy for construction quality and economic use. Larger mean areas suggest permanent, purpose-built structures (warehouses, schools, government offices) rather than informal dwellings. In the model, this is a per-buffer average rather than a total.

Total_Bldg_Proportion

Fraction of the buffer area covered by building footprints. Captures urban density independently of buffer size. Dense urban cores have high proportions; agricultural and forested areas near zero.



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Bldg_Density_per_km2

Number of buildings per square kilometer of buffer area. Normalizes building count by area to allow comparison across differently sized buffers.

Building Type Features (per type: residential, commercial, industrial, school, hospital)

Each building type has four sub-features:

{Type}_Count

Number of buildings of this type in the buffer. Provides a direct count of typed facilities. School count and hospital count in particular are strong indicators of public service investment.

{Type}_TotalArea (m²)

Combined footprint area of all buildings of this type.

**{Type}_MeanArea (m²
per-buffer average)**

Average footprint of buildings of this type. Larger mean school footprint areas indicate more substantial school infrastructure, associated with higher government investment and better-resourced communities. This is one of the strongest building-level predictors in the model.

{Type}_Proportion

Fraction of buffer area covered by this building type.



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4 OSM Point-of-Interest (POI) Features

Source: OpenStreetMap POIs layer.

POI features count service facilities within the prediction point buffer. Service facility density captures commercial activity, public investment, and access to essential services, all of which are correlated with household wealth.

Total_POI_Count	Total count of all service facilities across all categories below. The aggregate indicator of economic footprint and service density. Log1p-transformed before model input due to right-skew. The second-strongest static predictor in the study.
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Financial Services

POI_bank_Count	Number of bank branches.
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POI_atm_Count	Number of ATMs.
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Food and Retail

POI_restaurant_Count	Number of restaurants.
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POI_fast_food_Count	Number of fast food establishments.
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POI_convenience_Count	Number of convenience stores.
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POI_market_Count	Number of public markets or market halls.
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Health and Education

POI_hospital_Count	Number of hospitals, health centers, and clinics.
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POI_school_Count	Number of schools at all levels.
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POI_pharmacy_Count	Number of pharmacies.
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Government and Public Services

POI_government_Count	Number of government offices (municipal halls, LGU offices, DILG facilities).
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POI_police_Count	Number of police stations.
POI_post_office_Count	Number of post offices.
POI_community_Count	Number of community centers, barangay halls, and multi-purpose facilities.
POI_hotel_Count	Number of hotels and guesthouses.
Sports and Recreation	
POI_sports_Count	Number of sports facilities.

5 OSM Landuse Features

Source: OpenStreetMap landuse layer.

Unit: square meters (m², reported as km² in the dashboard)

Landuse polygon areas capture the macro-level land use composition around each prediction point. These are the strongest static predictors in the model after road length, because landuse classification directly reflects the economic character of an area's surrounding environment.

LU_Agricultural_m2	Total area of agricultural land use within the buffer. Covers cropland, farmland, paddy fields, and plantation areas. The strongest single static predictor in the study. High agricultural area is associated with lower predicted wealth, reflecting subsistence farming contexts typical of the five poorest provinces. Log1p-transformed.
LU_Forest_m2	Total area of forest cover within the buffer. The third-strongest static predictor. Predominantly associated with negative SHAP values, reflecting that forest-adjacent communities in Kalinga, Benguet, and BARMM island provinces tend to be geographically isolated with limited road access and low economic activity. Log1p-transformed.



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LU_Residential_m2

Total area mapped as residential land use. Captures formal residential zoning distinct from individual building footprints. Higher values indicate planned or semi-formal settlement patterns.

LU_Commercial_m2

Total area of commercial zones. Presence of mapped commercial land use indicates formal economic zoning, associated with higher wealth areas.

LU_Industrial_m2

Total area of industrial zones. Presence of industrial land use indicates manufacturing or processing activity, associated with employment and higher regional wealth.